Amazon ML Summer School 2025

Assessment – Afternoon Shift (12:00 PM)

Curated for practice by Sanjay Choudhary

Section 1: MCQ Section

Note to Student:
Please solve the following questions carefully. Evaluate all options and select the most appropriate answer. No solutions are provided in this test booklet to simulate a real exam environment.

Question 1: A discrete random variable X takes values 0, 1, 2, 3 with $P(X = 0) = 0.1$, $P(X = 1) = 0.2$, $P(X = 2) = 0.3$, and $P(X = 3) = 0.4$. Find $Var(X)$.

A) 0.9
B) 0.8
C) 1.0
D) 1.1

Question 2: Three fair dice are rolled. What is the probability that the sum is exactly 10?

A) $\frac{6}{216}$

B) $\frac{10}{216}$

C) $\frac{12}{216}$

D) $\frac{9}{216}$

Question 3: Determine whether the series $\sum_{n=1}^{\infty} \frac{n}{n^3+1}$ converges.

A) Diverges by Integral Test

B) Converges absolutely by Ratio Test

C) Converges by Limit Comparison with $\frac{1}{n^2}$

D) Diverges by comparison with $\frac{1}{n}$

Question 4: You deployed a sentiment analysis model that initially worked well, but its performance degraded over the last month. What is the best next step?

A) Switch to a bigger model

B) Re-train the model immediately

C) Collect recent user data and evaluate model drift

D) Increase the learning rate

Question 5: Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. What is the inverse of A?

A) $\begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$

B) $\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$

C) $\begin{bmatrix} -1 & 2 \\ 3 & -4 \end{bmatrix}$

D) $\begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$

Question 6: You accidentally include the target variable in the scaling features. What happens?

A) You applied PCA incorrectly

B) The model generalized exceptionally well

C) The test set was corrupted

D) Data leakage inflated model performance

Question 7: Let $f(x) = x^4 - 4x^3 + 5$. What type of critical point occurs at $x = 3$?

A) Point of inflection

B) No critical point

C) Local minimum

D) Local maximum

Question 8: A fair coin is tossed 6 times. What is the probability of getting exactly 4 heads?

A) $\frac{3}{8}$

B) $\frac{15}{64}$

C) $\frac{1}{2}$

D) $\frac{5}{16}$

Question 9: In reinforcement learning, the Bellman Equation is used to:

A) Approximate kernel similarity

B) Compute gradients for backpropagation

C) Calculate the entropy of policy

D) Evaluate the value of a policy

Question 10: In K-Means clustering, which of the following is a limitation that can significantly affect the algorithm's performance?

A) It assumes clusters of similar density

B) It can handle non-spherical clusters efficiently

C) It automatically selects the optimal number of clusters

D) It is robust to outliers

Question 11: If a population has mean $\mu = 100$ and standard deviation $\sigma = 20$, what is the standard deviation of the sampling distribution of the sample mean for samples of size 25?

A) 10 B) 20
C) 5 D) 4

Question 12: A manufacturer claims that the average weight of a product is 500g. A sample of 36 items has a mean weight of 492g and standard deviation of 18g. What is the Z-score?

A) -2.67 B) 1.33
C) 2.67 D) -1.33

Question 13: Let X be a continuous random variable uniformly distributed over the interval $[2, 8]$. What is the probability that X lies between 4 and 6?

A) $\frac{2}{3}$

B) $\frac{1}{6}$

C) $\frac{1}{2}$

D) $\frac{1}{3}$

Section 2: Coding Section

TEST INSTRUCTIONS

- **GENERAL**
This is a timed test. If you feel the question is unclear, try out different inputs and check the expected output for each of them.
- **CODE IMPLEMENTATION**
You only need to implement the given function. Do not read input; instead, use the arguments to the function. Do not print the output; instead, return values as specified.
- **PLAGIARISM**
Your code submission will be checked for plagiarism. To prevent plagiarism, we discourage switching tabs or windows during the test.

Q1. Covid-20?

Problem Description
A lab was trying to synthesize a new virus. They had 1 cell of that virus in a vial on day 1. The virus starts giving birth to a new cell every day after B days and it also dies after C days. On a new day, the cell will die (if it is supposed to die) before it gives birth to new cells. Find the number of virus cells in the vial after A days.
Note: Since the answer can be very large, output the answer modulo $10^9 + 7$.

Problem Constraints
$1 \leq A \leq 10^5$
$1 \leq B \leq A$
$B < C \leq A$

Input Format
The first argument is the integer A.
The second argument is the integer B.
The third argument is the integer C.

Example Input
Input 1:
$A = 5$, $B = 2$, $C = 3$
Input 2:
$A = 5$, $B = 1$, $C = 5$

Example Output
Output 1: 2
Output 2: 16

Example Explanation
Explanation 1:
The number of cells in the vial from day 1 to day 5 are $[1, 1, 2, 1, 2]$.
Explanation 2:
The number of cells in the vial from day 1 to day 5 are $[1, 2, 4, 8, 16]$.

Q2. Library Book Tracking

Problem Description
You are responsible for tracking the availability of books in a library that undergoes continuous updates. You have two integer arrays `A` and `B` of length `N`. Each element in `A` represents a unique book ID, and the corresponding element in `B` indicates how many copies of that book should be added to or removed from the library's collection at each step.

- **Addition of Books:** If `B[i]` is positive, it means `B[i]` copies of the book with ID `A[i]` are added to the library at step `i`.
- **Removal of Books:** If `B[i]` is negative, it means `-B[i]` copies of the book with ID `A[i]` are removed from the library at step `i`.

Return an array `result` of length `N`, where `result[i]` represents the count of the highest number of copies available for any single book in the library after the i^{th} step.
If the library has no books at any step, `result[i]` should be `0` for that step.

Problem Constraints
$1 \leq$ A.length == B.length $\leq 10^5$
$1 \leq A[i] \leq 10^5$
$-10^5 \leq B[i] \leq 10^5$
$B[i] \neq 0$
The input is generated such that the occurrences of an ID will not be negative in any step.

Input Format
First Argument is an Integer Array A, denoting the Book IDs.
Second Argument is an Integer Array B, denoting the count of books removed or added.

Example Input
Input 1: $A = [2, 3, 2, 1]$
$B = [3, 2, -3, 1]$
Input 2: $A = [5, 5, 3]$
$B = [2, -2, 1]$

Example Output
Output 1: $[3, 3, 2, 2]$
Output 2: $[2, 0, 1]$

Example Explanation
Explanation 1:
- **After step 0:** We have 3 copies of the book with ID 2. So `result[0] = 3`.
- **After step 1:** We have 3 copies of the book with ID 2 and 2 copies of the book with ID 3. The maximum copies for any single book is 3. So `result[1] = 3`.
- **After step 2:** 3 copies of the book with ID 2 are removed. We are left with 2 copies of the book with ID 3. The maximum is 2. So `result[2] = 2`.
- **After step 3:** 1 copy of the book with ID 1 is added. We have 2 copies of the book with ID 3 and 1 copy of the book with ID 1. The maximum is 2. So `result[3] = 2`.

Explanation 2:
- **After step 0:** We have 2 copies of the book with ID 5. So `result[0] = 2`.
- **After step 1:** 2 copies of the book with ID 5 are removed. There are no copies of any books left in the library. So `result[1] = 0`.
- **After step 2:** We have 1 copy of the book with ID 3. So `result[2] = 1`.

End of Assessment

These are the **real assessment questions from the Amazon ML Summer School 2025 (Afternoon Shift 12:00 PM)**.

For more content, solved papers, and guidance, feel free to connect or visit my GitHub repository: **github.com/cu-sanjay**
If you find this document helpful, please star the repository and follow the account.